Online Retail II
Data Analysis Report

Transaction Patterns, Geographic Markets, and Product Performance

Project objective
Identify sales-related patterns in the Online Retail II dataset and translate the findings into practical questions and opportunities for commercial improvement.

Analytical scope
This report summarizes data preparation, monthly transaction patterns, geographic distribution, and product-level results. It uses previously generated datasets and visualizations.

Measurement note
Line amount is calculated as quantity multiplied by unit price. Because the source contains cancellations, adjustments, and exceptional transactions, line amounts should not automatically be interpreted as verified net revenue or profit.

Prepared as a data analyst portfolio project

1. Data and Methodology

The original workbook contains two yearly sheets with an overlapping date period in December 2010. During preparation, 22,523 matching transaction rows were identified in the overlap and removed from the earlier sheet's overlapping period in the analysis dataset. The original workbook was preserved.

Dataset stage	Rows	Purpose
Combined standardized data	1,067,371	Standardized source records and calculated line amounts
Prepared analysis data	1,044,848	Analysis dataset after handling the identified overlap

Data preparation and quality considerations

• The source contains missing customer IDs and product descriptions.

• Cancellation records and negative quantities were retained for investigation.

• Some stock codes represent postage, fees, or adjustments rather than physical products.

• Some stock codes have multiple descriptions; descriptions were not automatically corrected.

• Exceptional positive and negative transactions were investigated rather than silently deleted.

The separate Online Retail II Data Quality Report documents the detailed quality checks, exceptional cases, and preparation decisions.

2. Monthly Transaction Patterns

Monthly results show changes in transaction activity and line amounts over time. November has high activity in both observed years. This is a pattern worth investigating, but two years of observations are not sufficient to establish a stable seasonal effect.

2010-11: 78,015 transaction lines; 3,669 unique invoices; line amount total 1,422,654.64.

2011-11: 84,711 transaction lines; 3,462 unique invoices; line amount total 1,461,756.25.

2011-12: 25,526 transaction lines; 1,015 unique invoices; line amount total 433,686.01.

December 2011 is incomplete, with source records ending on December 9. It should not be compared with complete months as if it represented a full month. Large cancellation-related amounts also affect some monthly totals.

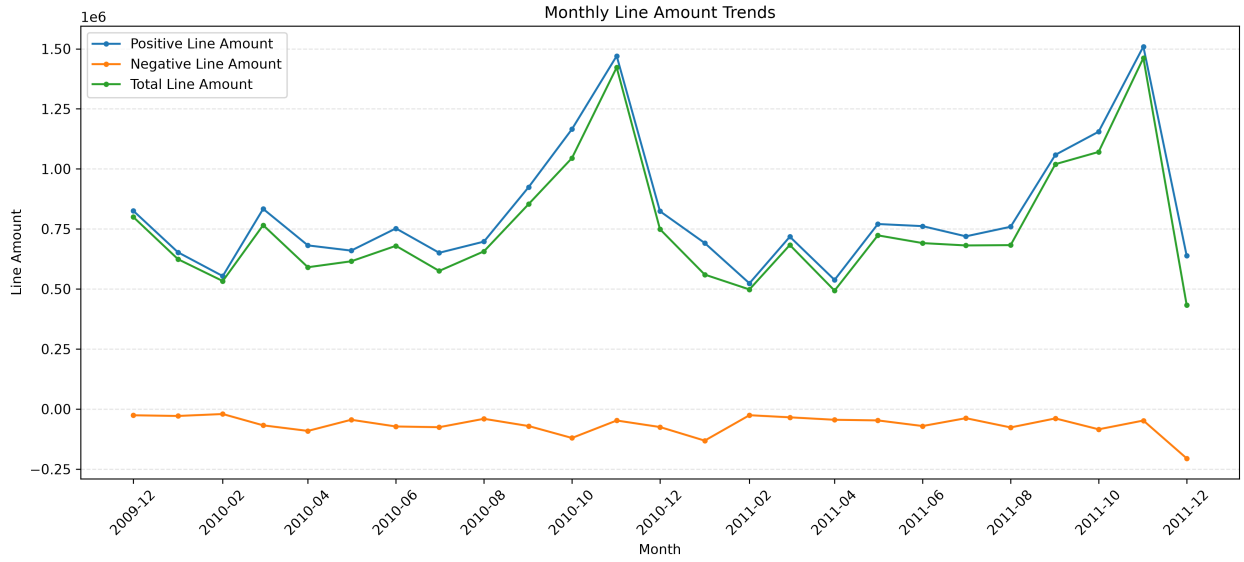


Figure 1. Monthly line amount totals. Interpret peaks alongside cancellations, exceptional transactions, and incomplete periods.

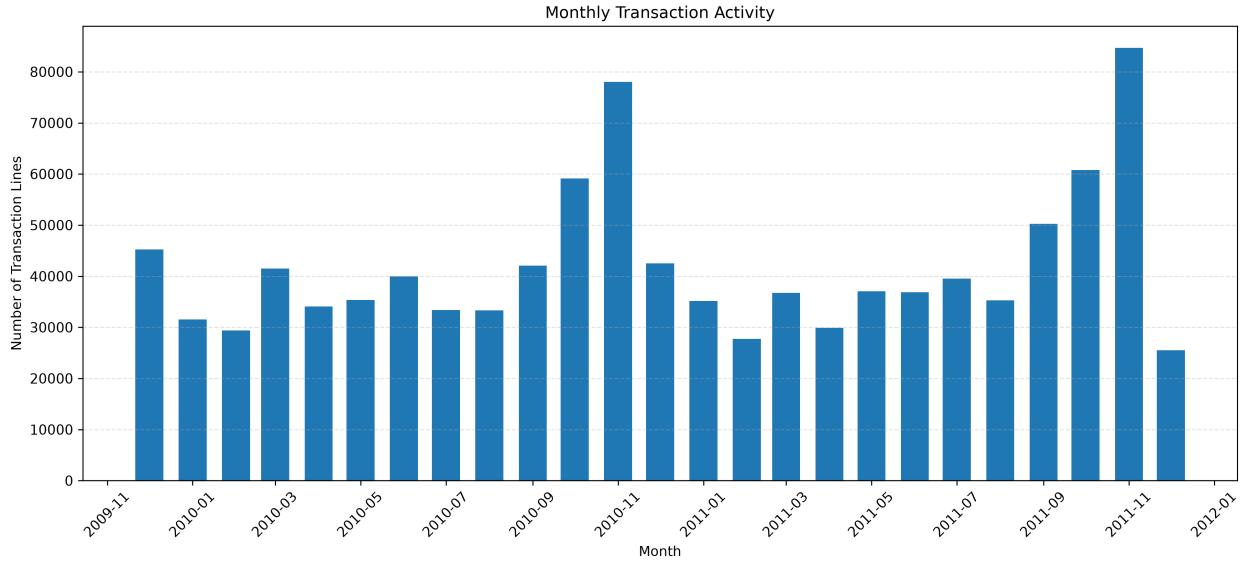


Figure 2. Monthly transaction activity.

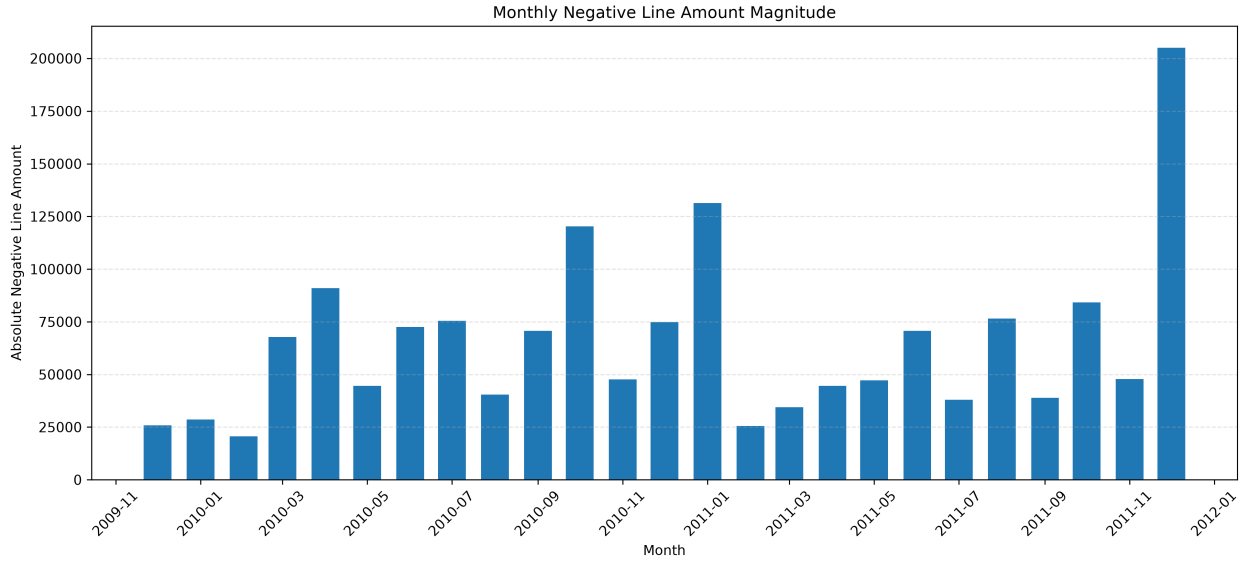


Figure 3. Monthly negative line amounts, highlighting periods that warrant further review.

3. Geographic Distribution

The United Kingdom accounts for 959,983 transaction lines, approximately 91.88% of the country-level transaction-line total.

EIRE and Germany are among the next largest markets by transaction-line count, while EIRE and the Netherlands rank highly by positive line amount. The difference between these rankings suggests that transaction frequency and positive line amount provide complementary views of market activity.

The concentration in the UK suggests that the observed business activity is heavily dependent on that market. International markets may warrant further investigation, but expansion decisions would require evidence on customer acquisition, delivery costs, margins, and market potential.

Country	Transaction lines	Positive line amount
United Kingdom	959,983	17,466,269.91
EIRE	17,689	659,149.00
Germany	17,363	425,578.38
France	14,059	350,654.18
Netherlands	5,138	554,039.74

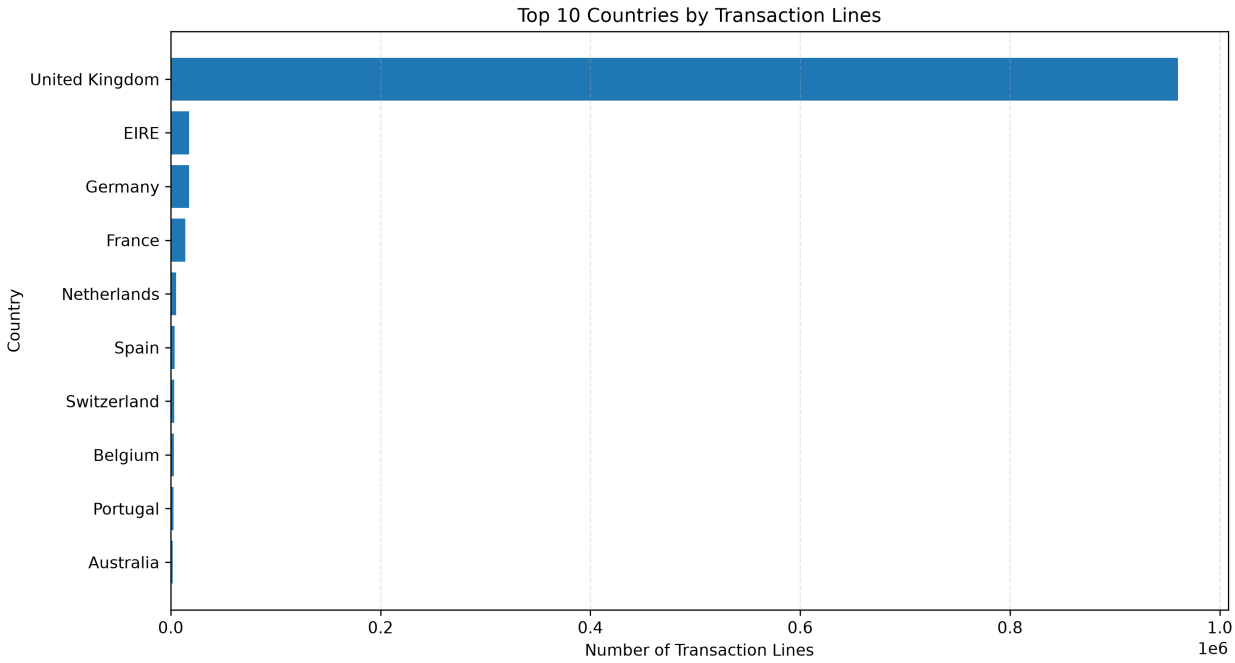


Figure 4. Countries ranked by transaction-line count.

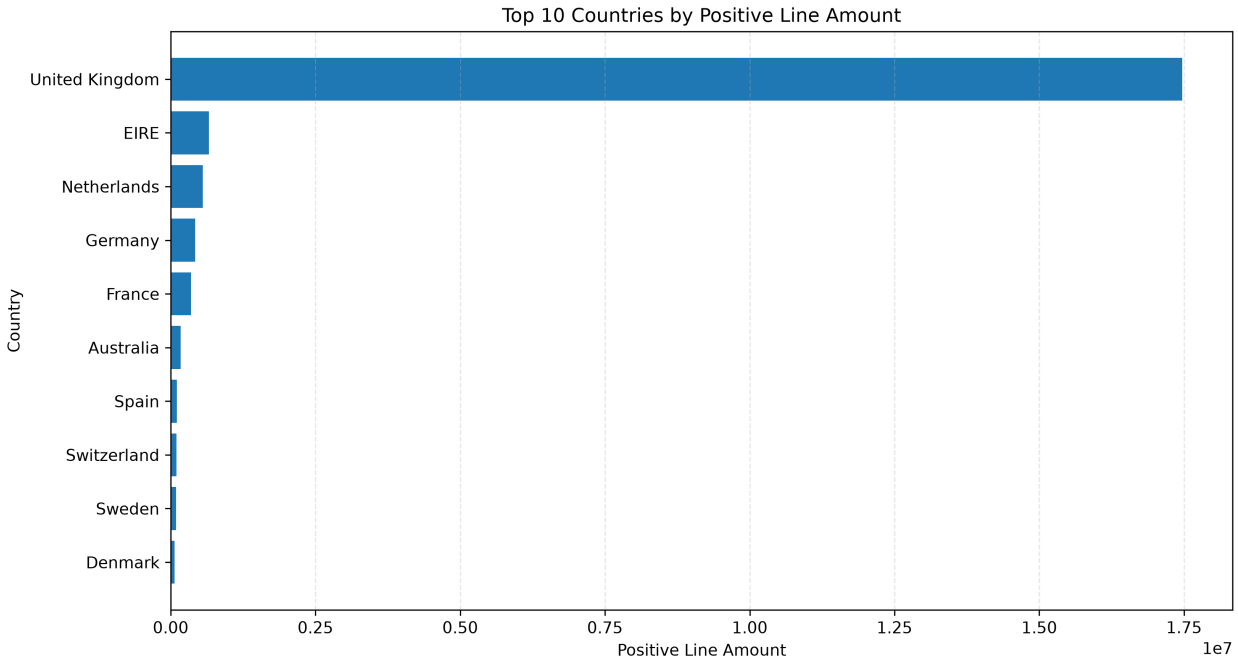


Figure 5. Countries ranked by positive line amount.

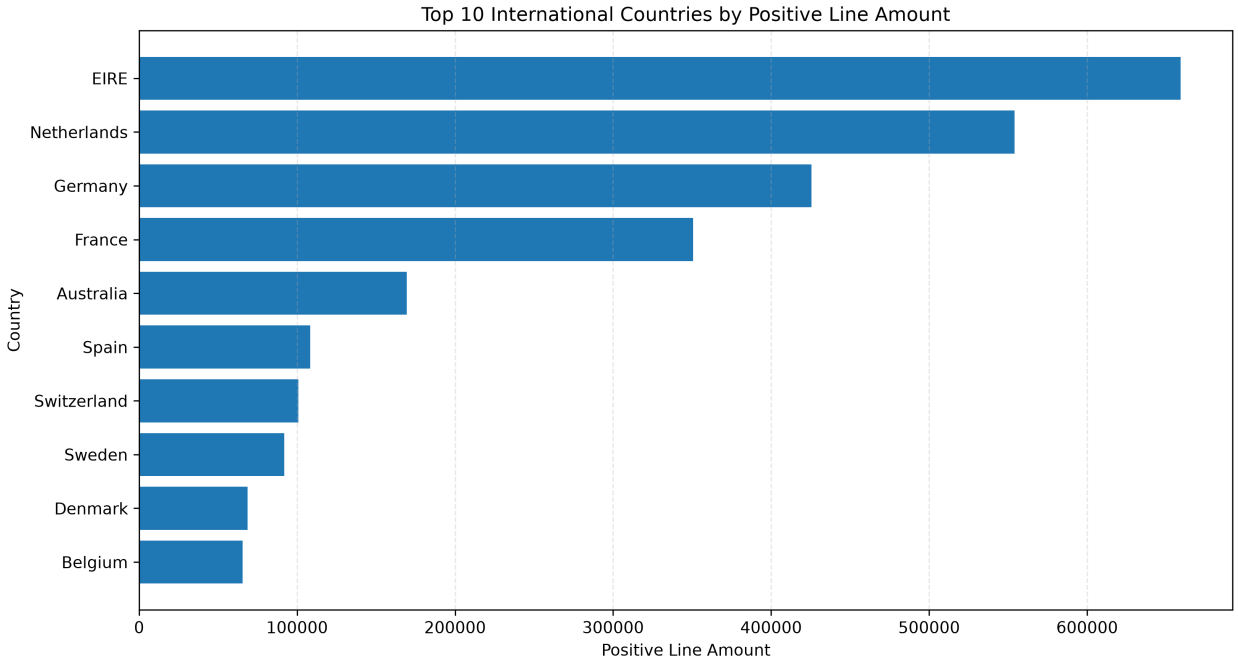


Figure 6. Leading international markets by positive line amount.

4. Product-Level Findings

Product comparisons exclude known non-standard stock codes and two previously investigated exceptional product codes. This filter is used only for comparative rankings; it does not remove records from the underlying prepared dataset.

Top 10 Products by Positive Line Amount

Stock code	Product	Lines	Positive amount
22423	REGENCY CAKESTAND 3 TIER	4,327	331,084.27
85123A	WHITE HANGING HEART T-LIGHT HOLDER	5,711	258,065.91
85099B	JUMBO BAG RED RETROSPOT	4,147	180,888.03
47566	PARTY BUNTING	2,744	148,395.98
84879	ASSORTED COLOUR BIRD ORNAMENT	2,896	129,681.08
22086	PAPER CHAIN KIT 50'S CHRISTMAS	2,111	117,884.19
79321	CHILLI LIGHTS	1,184	80,873.23
22197	SMALL POPCORN HOLDER	2,479	79,552.50
22386	JUMBO BAG PINK POLKADOT	2,312	76,385.20
20725	LUNCH BAG RED RETROSPOT	3,208	71,396.14

Top 10 Products by Transaction-Line Frequency

Stock code	Product	Lines	Unique invoices
85123A	WHITE HANGING HEART T-LIGHT HOLDER	5,711	5,507
22423	REGENCY CAKESTAND 3 TIER	4,327	4,272
85099B	JUMBO BAG RED RETROSPOT	4,147	4,078
21212	PACK OF 72 RETROSPOT CAKE CASES	3,260	3,144
20725	LUNCH BAG RED RETROSPOT	3,208	3,135
84879	ASSORTED COLOUR BIRD ORNAMENT	2,896	2,829
47566	PARTY BUNTING	2,744	2,702
21232	STRAWBERRY CERAMIC TRINKET BOX	2,709	2,618
22383	LUNCH BAG SUKI DESIGN	2,503	2,431
20727	LUNCH BAG BLACK SKULL.	2,490	2,396

Regency Cakestand 3 Tier has the highest positive line amount in the filtered comparison. White Hanging Heart T-Light Holder has the highest transaction-line count and appears in the most unique invoices among the listed products. Jumbo Bag Red Retrospot also ranks highly on both measures.

These findings show that product frequency and positive line amount are different indicators. Neither metric establishes profitability, and both must be interpreted alongside returns, cancellations, and the limitations of the source data.

5. Business Opportunities

Review product availability

Use frequently purchased products as candidates for stock availability and replenishment reviews, subject to inventory and supply data.

Investigate international markets

Compare selected international markets using customer counts, order values, delivery costs, and margins before recommending expansion.

Monitor cancellations and adjustments

Report exceptional cancellations separately so unusual transactions do not distort product or monthly comparisons.

Improve product data governance

Review stock codes with conflicting or missing descriptions and distinguish physical products from charges and adjustments.

Validate seasonal hypotheses

Analyze additional complete years before drawing conclusions about seasonality or making calendar-based sales forecasts.

6. Limitations

• The dataset represents historical activity and may not reflect current behavior.

• December 2011 is incomplete.

• Missing customer IDs limit customer-level analysis.

• Line amounts do not establish profit or verified net revenue.

• Observed rankings and patterns do not prove causation or future demand.

• Business recommendations require additional operational and financial data.

7. Conclusion

The analysis identifies a highly concentrated UK market, months with elevated transaction activity, and products that stand out by frequency or positive line amount. The main value of the project is the combination of data preparation, quality checks, descriptive analysis, and cautious business interpretation. Further operational and financial data would be needed to validate the proposed commercial opportunities.

8. Project Outputs

• 03_Results/monthly_sales_overview.csv

• 03_Results/country_sales_overview.csv

• 03_Results/product_sales_overview.csv

• 03_Results/Graficas/

• 04_Documentation/Online_Retail_II_Data_Quality_Report.pdf